

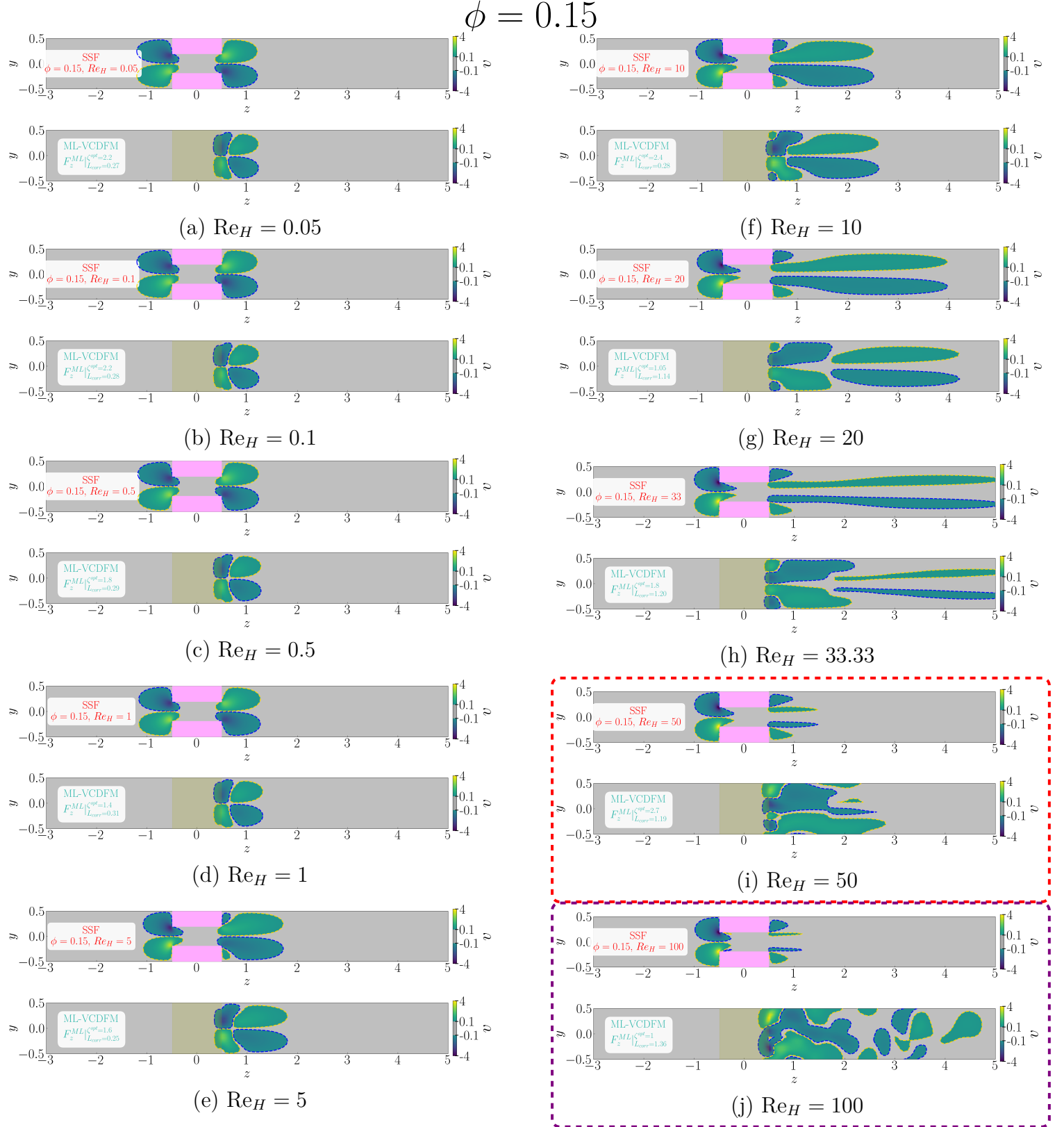


Figure D.8: Transverse velocity component v in the streamwise (z, y) -plane for the SRG at $\phi = 0.15$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the ML-VCDFM reconstruction using F_z^{ML} . The red dashed rectangle marks $Re_H = 50$, where noticeable differences from the SSF indicate a loss of ML-VCDFM accuracy within the training parameter range. The violet dashed rectangle marks the extrapolation case $Re_H = 100$, where significant differences from the SSF place the reconstruction outside its effective validity range.

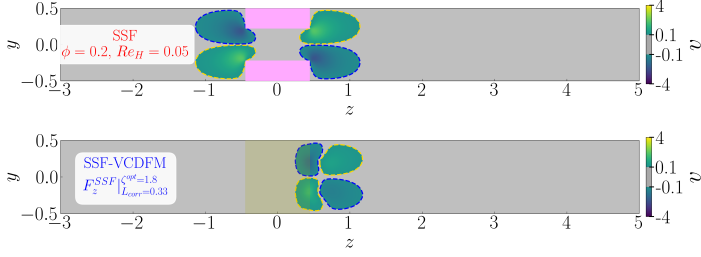
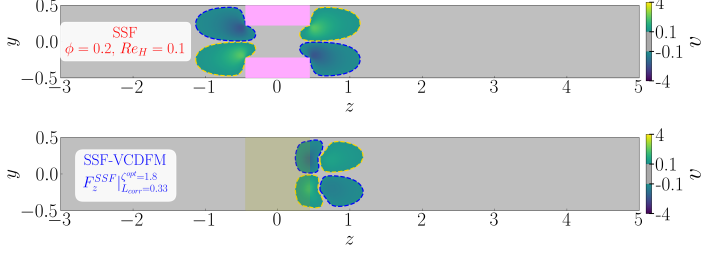
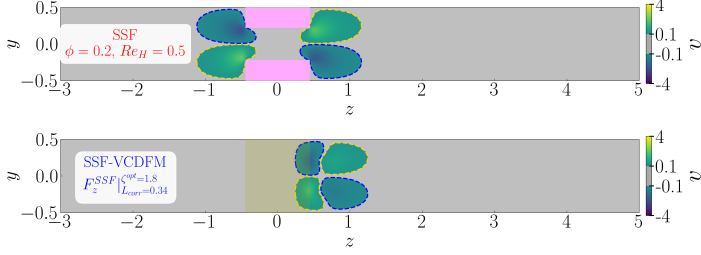
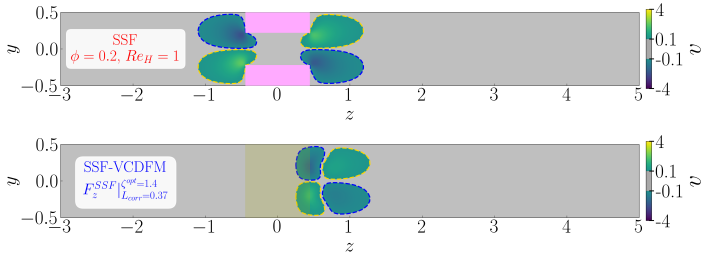
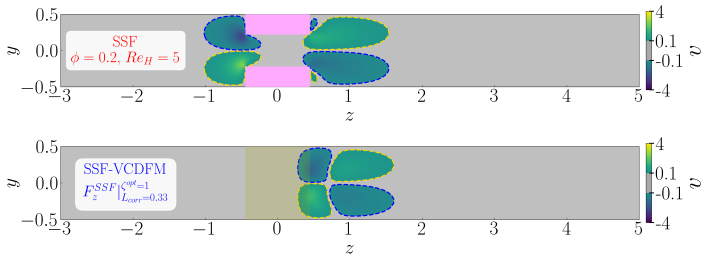
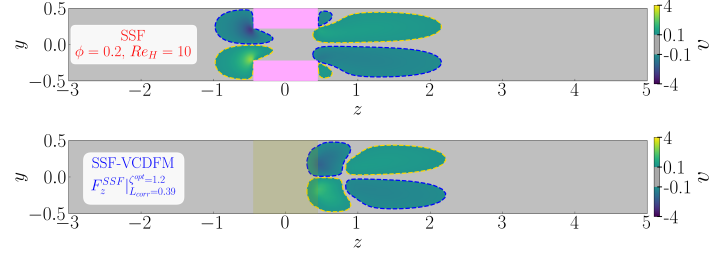
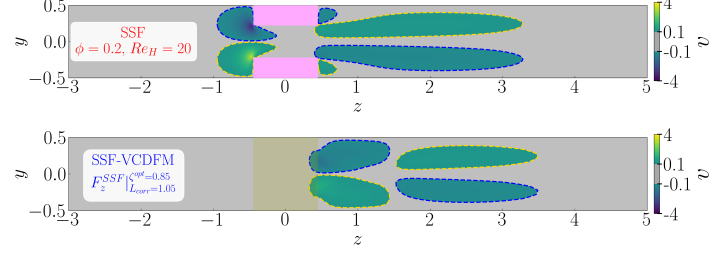
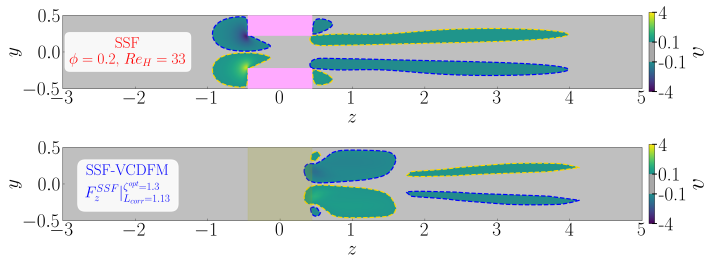
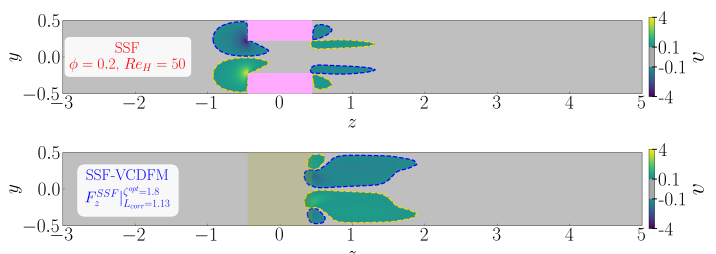
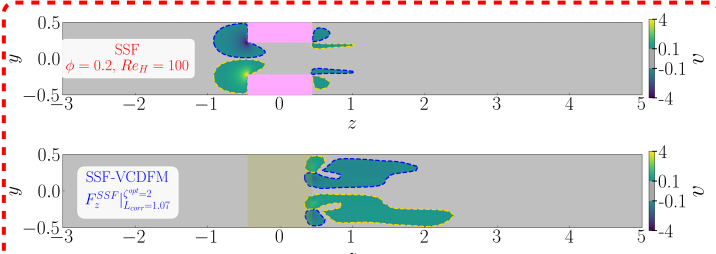
$\phi = 0.20$

(a) $Re_H = 0.05$

(b) $Re_H = 0.1$

(c) $Re_H = 0.5$

(d) $Re_H = 1$

(e) $Re_H = 5$

(f) $Re_H = 10$

(g) $Re_H = 20$

(h) $Re_H = 33.33$

(i) $Re_H = 50$

(j) $Re_H = 100$

Figure D.9: Transverse velocity component v in the streamwise (z,y) -plane for the SRG at $\phi = 0.20$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the SSF-VCDFM reconstruction using F_z^{SSF} . The red dashed rectangle marks $Re_H = 100$, where noticeable differences between the SSF-VCDFM and SSF indicate a loss of predictive accuracy.

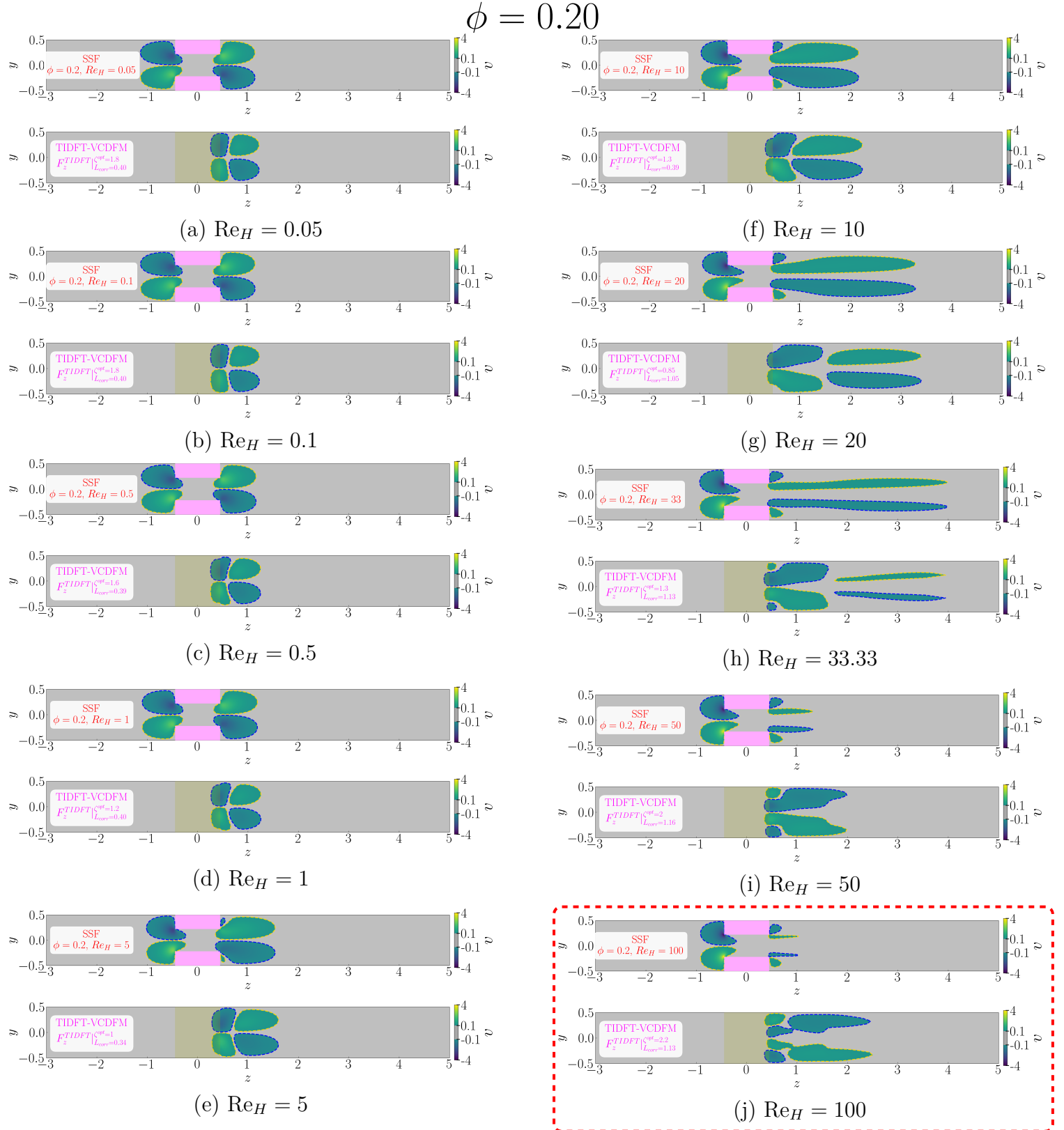


Figure D.10: Transverse velocity component v in the streamwise (z, y) -plane for the SRG at $\phi = 0.20$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the TIDFT-VCDFM reconstruction using F_z^{TIDFT} . The red dashed rectangle marks $Re_H = 100$, where noticeable differences between the TIDFT-VCDFM and SSF indicate a loss of predictive accuracy.

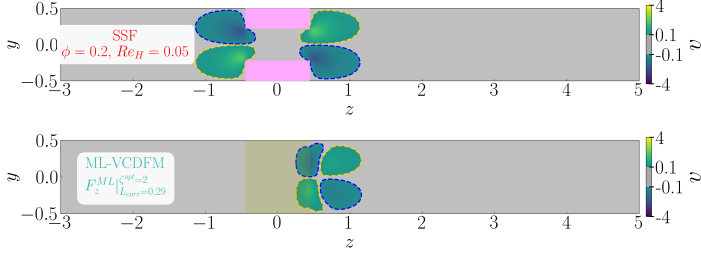
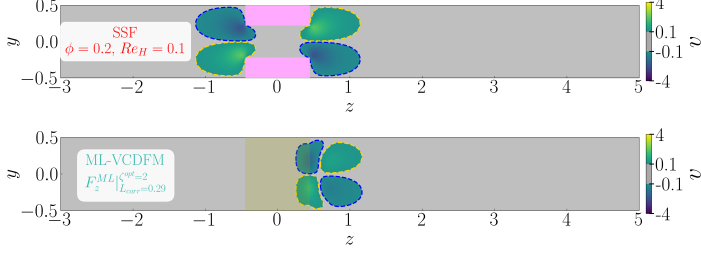
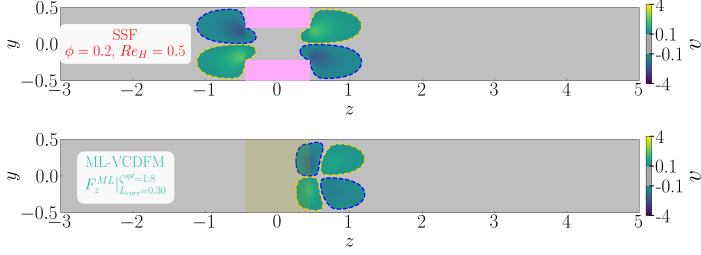
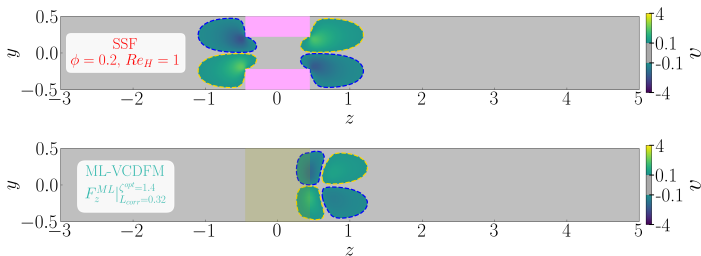
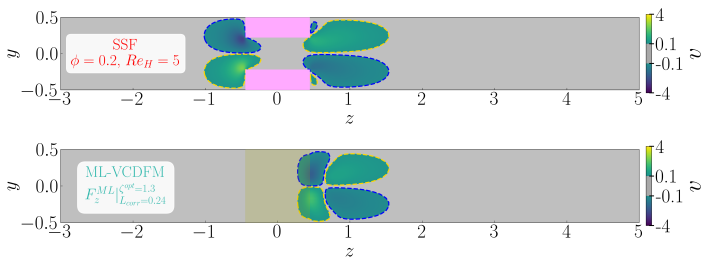
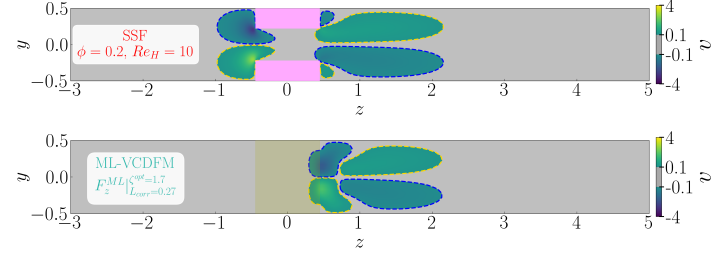
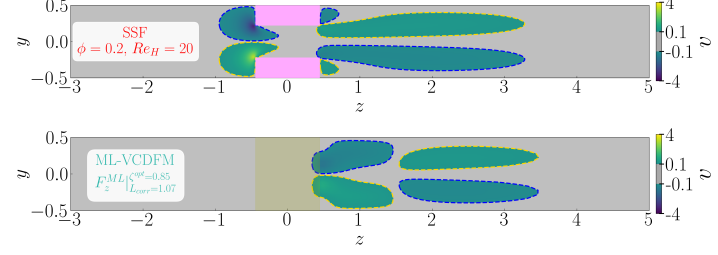
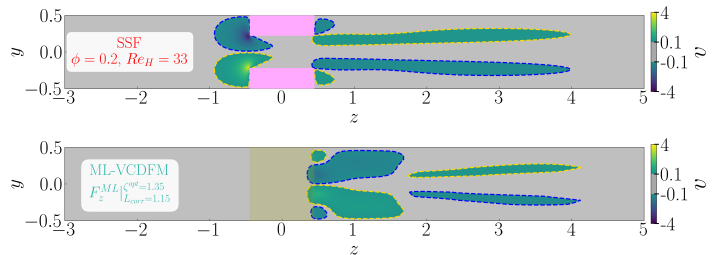
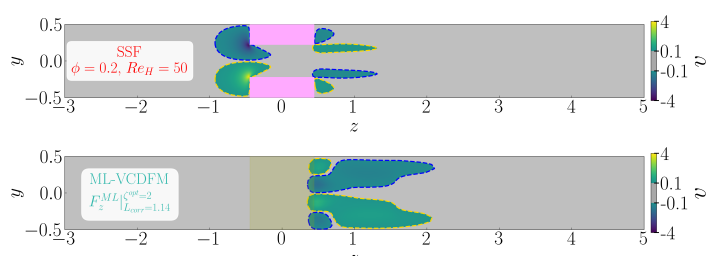
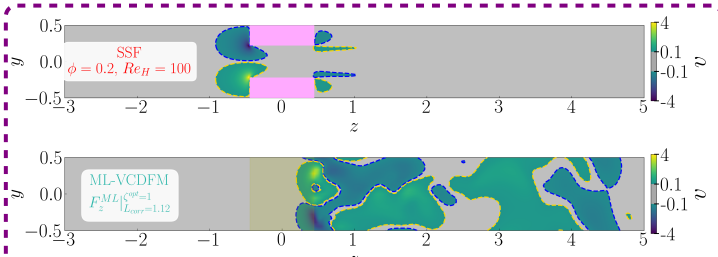
$\phi = 0.20$

(a) $Re_H = 0.05$

(b) $Re_H = 0.1$

(c) $Re_H = 0.5$

(d) $Re_H = 1$

(e) $Re_H = 5$

(f) $Re_H = 10$

(g) $Re_H = 20$

(h) $Re_H = 33.33$

(i) $Re_H = 50$

(j) $Re_H = 100$

Figure D.11: Transverse velocity component v in the streamwise (z,y) -plane for the SRG at $\phi = 0.20$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the ML-VCDFM reconstruction using F_z^{ML} . The violet dashed rectangle marks the extrapolation case $Re_H = 100$, where significant differences from the SSF place the ML-VCDFM reconstruction outside its effective validity range.

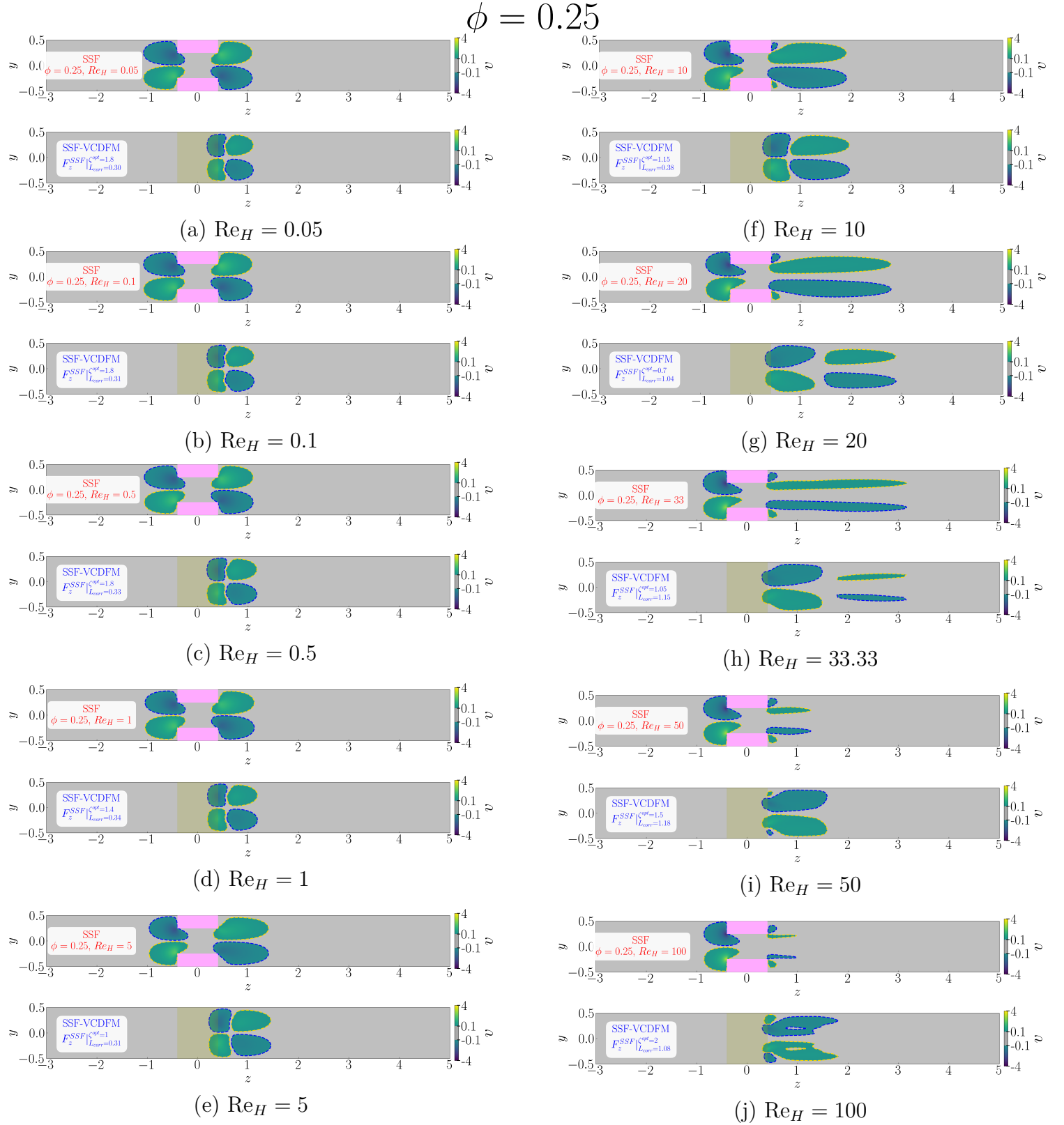


Figure D.12: Transverse velocity component v in the streamwise (z, y) -plane for the SRG at $\phi = 0.25$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the SSF-VCDFM reconstruction using F_z^{SSF} .